
Production Scalability Architecture: Frontend, Microservices, Distributed Storage, Geo-Search, and Deployment
1. CLIENT TIER (Cross-Platform)
Web App (Next.js / React ISR)
iOS & Android Apps (React Native)
Host Management Portal & Admin Console
2. EDGE & CDN NETWORK (Global Latency < 30ms)
Cloudflare / AWS CloudFront
DDoS Protection & WAF Rules
Edge SSR Cache (Stale-While-Revalidate)
GeoDNS Routing & SSL Term.
3. API GATEWAY & SECURITY LAYER
Kong / AWS API Gateway
OAuth2 / JWT / Auth0 Service
Rate Limiter (Redis Token Bucket)
OpenTelemetry Tracing & Logs
4. CORE MICROSERVICES DOMAIN (Containerized on Kubernetes / EKS)
Listing Service
• Property Details
• 50+ Amenities Modal
• Multi-lang Translations
Booking Engine
• Double-Booking Lock
• 2-Phase Hold (10m TTL)
• Calendar Availability
Search & Discovery
• Geo-Bounding Polygon
• Dynamic Price Filters
• Instant Autocomplete
Reviews & Ratings
• 7 Category Breakdown
• Guest Favourite Score
• Sentiment Aggregation
Payments & Ledger
• Stripe / Gateways
• Host Automated Payouts
• Idempotent Webhooks
Realtime & Notifs
• Host Messaging
• Push / SMS Dispatch
• WebSocket Streams
EVENT STREAMING & ASYNC WORKERS (Apache Kafka / AWS MSK)
[Topic: ListingEvents]
[Topic: BookingCreated]
[Topic: PaymentSucceeded]
[Topic: ReviewPublished]
5. DISTRIBUTED STORAGE, CACHE, & SEARCH TIER
PostgreSQL / Aurora (ACID)
• Users, Listings, Bookings
• Multi-AZ Active-Passive
• Global Read Replicas
Redis Cluster (ElastiCache)
• Calendar Availability Bitmaps
• Distributed Redlock
• Session & Cart Hold State
OpenSearch / Elasticsearch
• Geo-point Spatial Indexing
• 50+ Amenity Filtering Index
• Realtime Kafka Sync Consumer
AWS S3 / R2 + Lambda
• High-Res Property Photos
• Auto WebP / AVIF Compression
• Signed URLs for Host Uploads
6. DEPLOYMENT, CI/CD, & OBSERVABILITY PIPELINE
GitHub Actions / ArgoCD GitOps
Kubernetes EKS + Karpenter Autoscaling
Prometheus + Grafana Dashboards
Terraform Multi-Region IaC